Construction of the Real Numbers from the Rational Numbers

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Contents

Contents 1

1 Introduction 2

2 Historical Background 2

3 Preliminaries 2
3.1 Definitions . . . 2
3.2 Motivation . . . 3

4 Dedekind Cuts 3
4.1 Definitions . . . 3
4.2 Important Lemmas . . . 4
4.3 Operations . . . 5
4.3.1 Addition . . . 5
4.3.2 Multiplication . . . 6
4.4 Ring Axioms . . . 8
4.5 Order Axioms . . . 9
4.6 Least Upper Bound . . . 10

5 Real Numbers 11
5.1 Construction . . . 11
5.2 Uniqueness . . . 12

6 Cauchy Sequences 13
6.1 Completeness . . . 13

7 Conclusion 14

1. Introduction

The goal of this writeup is to construct the real numbers from the rational numbers. Although $\mathbb{Q}$ is an ordered field, it does not satisfy the least upper bound property. For example, the set of rational numbers whose squares are less than 2 is bounded above in $\mathbb{Q}$, but it has no least upper bound in $\mathbb{Q}$. This shows that the rational numbers contain gaps.

To fill these gaps, we define real numbers using Dedekind cuts. A Dedekind cut is a subset of $\mathbb{Q}$ that represents all rational numbers lying to the left of a number. This allows irrational numbers to be described using only rational numbers. After defining these cuts, we prove that they form an ordered field and satisfy the least upper bound property.

2. Historical Background

The need for a precise construction of the real numbers became especially important in the nineteenth century, when mathematicians wanted to make calculus more rigorous. Earlier, real numbers were often understood geometrically as points on a line, but this did not give a purely arithmetic explanation of irrational numbers.

Richard Dedekind gave one construction of the real numbers using cuts of the rational numbers. His idea was to describe a number by how it separates $\mathbb{Q}$ into rational numbers below it and rational numbers above it. In this construction, rational numbers appear naturally as cuts, and irrational numbers such as $\sqrt{2}$ arise from gaps in $\mathbb{Q}$.

Other constructions of $\mathbb{R}$ were also developed, including constructions using Cauchy sequences. Although these approaches look different, they produce the same kind of object: an ordered field with the least upper bound property. The Dedekind cut construction therefore gives a precise way to replace the informal number line with a rigorous mathematical structure.

3. Preliminaries

3.1. Definitions

Definition 3.1. Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. A number $L \in \mathbb{R}$ is called the *least upper bound* of S if $x \leq L$ for every $x \in S$, and if M is any upper bound of S, then $L \leq M$. The least upper bound of S is also called its *supremum* and is denoted by $\sup S$.

Definition 3.2. A ring F is an *ordered field* if F is a field that satisfies the order axioms.

The set of real numbers $\mathbb{R}$ should follow the following two axioms:

1. **The least upper bound property (LUB):** Every nonempty subset of $\mathbb{R}$ with an upper bound has a least upper bound.
2. $\mathbb{R}$ is an ordered field.

3.2. Motivation

We show that $x^2 = 2$ has a solution in $\mathbb{R}$.

Take $S = \{x \in \mathbb{R} : x^2 < 2\}$. This set is bounded by 2 and it is nonempty since 1 is in the set. By LUB, S has a least upper bound, $\alpha \in \mathbb{R}$. We now show that $\alpha^2 = 2$.

If $\alpha^2 < 2$, then $\alpha \in S$. We want to choose an $0 < \epsilon < \alpha$ such that $(\alpha + \epsilon)^2 = \alpha^2 + 2\alpha\epsilon + \epsilon^2 < 2$. Rearranging, $2\alpha\epsilon + \epsilon^2 < 2 - \alpha^2$. Also, notice that $2\alpha\epsilon + \epsilon^2 < 2\alpha\epsilon + \alpha\epsilon = 3\alpha\epsilon$. We choose an ϵ such that $3\alpha\epsilon < 2 - \alpha^2$. Thus, $\epsilon < \frac{2-\alpha^2}{3\alpha}$.

Now, $(\alpha + \epsilon)^2 < 2 \implies (\alpha + \epsilon) \in S$. This is a contradiction because the least upper bound by 3.1 has to be greater than or equal to all elements in the set.

If $\alpha^2 > 2$, then $\alpha \notin S$. We want to choose an $0 < \epsilon < \alpha$ such that $(\alpha - \epsilon)^2 = \alpha^2 - 2\alpha\epsilon + \epsilon^2 > 2$. Rearranging, $2\alpha\epsilon - \epsilon^2 < \alpha^2 - 2$. Also, notice that $2\alpha\epsilon - \epsilon^2 < 2\alpha\epsilon$. We take ϵ such that $2\alpha\epsilon < \alpha^2 - 2$. Thus, $\epsilon = \frac{\alpha^2-2}{2\alpha}$

Now, $(\alpha - \epsilon)^2 > 2$. This is a contradiction because the least upper bound by 3.1 has to be less than or equal to all other upper bounds.

Thus, $\alpha^2 = 2$.

Now, we show that $\mathbb{Q}$ does not satisfy LUB. If $\mathbb{Q}$ does have a LUB, we can find a solution to $x^2 = 2$ by considering S and repeating the same process. But, this is impossible since $\sqrt{2} \notin \mathbb{Q}$.

4. Dedekind Cuts

4.1. Definitions

Definition 4.1. A *Dedekind cut* is a subset $A \subseteq \mathbb{Q}$ satisfying the following three conditions:

(C1) $A \neq \varnothing$ and $A \neq \mathbb{Q}$.

(C2) If $a \in A$ and $r \in \mathbb{Q}$ with $r < a$, then $r \in A$.

(C3) If $a \in A$, then there exists $b \in A$ such that $a < b$.

Definition 4.2. For each rational number q, define the corresponding rational cut by

$$\widehat{q} = \{r \in \mathbb{Q} : r < q\}.$$

Thus,

$$\widehat{0} = \{r \in \mathbb{Q} : r < 0\} \text{ and } \widehat{1} = \{r \in \mathbb{Q} : r < 1\}$$

.

Proposition 4.3. *For every $q \in \mathbb{Q}$, the rational cut $\widehat{q}$ is a Dedekind cut.*

Proof. First, we show (C1). $\widehat{q}$ is nonempty as it contains the rational $q - 1$ since it is less than q. $\widehat{q} \neq \mathbb{Q}$ since it does not contain the rational $q + 1$ since it is greater than q.

(C2) is true by definition.

Now, we show (C3). Suppose we have $a \in \widehat{q}$. Let $b = \frac{a+q}{2}$. $a < b < q$.

Thus, $\widehat{q}$ is a Dedekind cut.

□

Definition 4.4. The order for Dedekind cuts is defined as $A < B$ if $A \subset B$ and $A \leq B$ if $A \subseteq B$.

4.2. Important Lemmas

Lemma 4.5. *If A is a Dedekind cut and $a \notin A$, then $\forall x \in A$, $x < a$.*

Proof. Let $x \in A$. We want to show that $x < a$.

FTSOC, suppose $x \geq a$. If $x = a$, then $a \in A$, which contradicts $a \notin A$. If $x > a$, then $a < x$. Since $x \in A$ and A is downward closed, we get $a \in A$, again a contradiction.

Therefore, $x < a$. □

Lemma 4.6. *If A is a Dedekind cut, it has an upper bound.*

Proof. By (C1), since $A \neq \mathbb{Q}$, there exists $a \notin A$. Thus, a is an upper bound by the lemma above. □

4.3. Operations

4.3.1. Addition

Definition 4.7. Suppose A and B are two Dedekind cuts. We define addition of the two cuts as $\{a+b : a \in A, b \in B\}$.

Definition 4.8. We define $-A = \{r \in \mathbb{Q} : \text{there exists } s \notin A \text{ such that } r < -s\}$.

Lemma 4.9. *If C is a Dedekind cut, then $-C$ is a Dedekind cut.*

Proof. We first consider (C1). Since $C \neq \mathbb{Q}$, there exists $c \notin C$. Then $-c-1 < -c$, so $-c-1 \in -C$. Thus, $-C$ is nonempty. Now, we show that $-C \neq \mathbb{Q}$. Since C is nonempty, take $d \in C$. We claim that $-d \notin -C$. If $-d \in -C$, then there exists some $r \notin C$ such that $-d < -r$. Thus, $r < d$. Since $d \in C$ and C is downward closed, this implies $r \in C$, a contradiction. Therefore $-d \notin -C$, so $-C \neq \mathbb{Q}$.

Now, we consider (C2). Suppose $x \in -C$ and $y < x$. Since $x \in -C$, there exists some $r \notin C$ such that $x < -r$. Since $y < x < -r$, we have $y < -r$. Therefore $y \in -C$. Thus, $-C$ is downward closed.

Finally, we consider (C3). Suppose $x \in -C$. Then there exists some $r \notin C$ such that $x < -r$. Choose $y = \frac{x-r}{2}$; so, $x < y < -r$. Then $y \in -C$ and $x < y$. Therefore $-C$ has no greatest element.

Thus, $-C$ is a Dedekind cut. □

Proposition 4.10. *The sum of two Dedekind cuts A and B is a Dedekind cut.*

Proof. We show that $S = A + B = \{a+b : a \in A, b \in B\}$. satisfies the Dedekind cut properties.

First, look at (C1). S is nonempty since A and B are nonempty because they are Dedekind cuts. Now, we show that $S \neq \mathbb{Q}$. Since $A \neq \mathbb{Q}$ and $B \neq \mathbb{Q}$, we can choose $a_1 \notin A$ and $b_1 \notin B$. Every element $a \in A$ is less than a_1 and every element $b \in B$ is less than b_1 since they are upper bounds of the sets. Thus, $a+b < a_1+b_1$. So, $a_1+b_1 \notin S$. Thus, $S \neq Q$ since it doesn't contain the rational number a_1+b_1.

Now, we look at (C2). FTSOC, suppose $a+b \in S$, $y < a+b$, and $y \notin S$. Because $y-b < a$, $y-b \in A$ by (C2). Therefore, $(y-b)+b = y \in S$, a contradiction. Thus, S is downward closed.

Finally, we look at (C3). FTSOC, suppose $a+b \in S$ and is the maximal element of the set. By (C3), there does not exist maximal elements in A and B since they are Dedekind cuts. So take some element $a' > a \in A$ and some element $b' > b \in B$. Thus, $a'+b' > a+b$. But by 4.3.1, $a'+b' \in S$. This is a contradiction. So, S does not have a maximal element.

Thus, S is a Dedekind cut. □

4.3.2. Multiplication

Definition 4.11. Let A and B be cuts with $A \geq \widehat{0}$ and $B \geq \widehat{0}$. Define

$$A \cdot B = \{r \in \mathbb{Q} : r < 0 \text{ or there exist } a \in A,\ b \in B \text{ with } a > 0,\ b > 0,\ r < ab\}.$$

For arbitrary cuts, define multiplication by the following sign rules:

$$\begin{aligned} A \cdot B &= \text{the product above}, && A \geq \widehat{0},\ B \geq \widehat{0}, \\ A \cdot B &= -\big((-A) \cdot B\big), && A < \widehat{0},\ B \geq \widehat{0}, \\ A \cdot B &= -\big(A \cdot (-B)\big), && A \geq \widehat{0},\ B < \widehat{0}, \\ A \cdot B &= (-A) \cdot (-B), && A < \widehat{0},\ B < \widehat{0}. \end{aligned}$$

Proposition 4.12. *The product of two Dedekind cuts A and B is a Dedekind cut.*

Proof. We assume $A \geq \widehat{0}, B \geq \widehat{0}$. Now, let's consider the three properties for $A \cdot B = S$.

First, we look at (C1). S is nonempty since we see it automatically includes all negative rationals. Now, we show $S \neq \mathbb{Q}$. A and B have an upper bound by 4.6. Thus, there exists $0 < a \notin A$ and $0 < b \notin B$. Now by 4.5, for positive $x \in A$, $x < a$, and positive $y \in B$, $y < b$. Thus, $xy < ab$. So, no element of S can be greater than xy. Thus, $S \neq \mathbb{Q}$.

Now, we look at (C2). If $x \in S$, then $x < 0$ or $x < ab$ for some $0 < a \in A$ and $0 < b \in B$. If $0 \leq r < x$ then $r < ab$ by transitivity, so $r \in S$ by definition. If $r < 0$, then it is in S by definition.

Now, we look at (C3). Suppose $x \in S$. We want to find some $y \in S$ such that $x < y$. If $x < 0$, take $y = \frac{x}{2}$. Since $x < 0$, we have $x < \frac{x}{2} < 0$, so $y < 0$. Therefore $y \in S$, and $x < y$. Now suppose $x \geq 0$. Since $x \in S$ and x is not negative, by the definition of multiplication there exist $a \in A$ and $b \in B$ with $a > 0$, $b > 0$, and $x < ab$. Take $y = \frac{x+ab}{2}$. Then $x < y < ab$, so $y \in S$. Also, $x < y$. Therefore, for every $x \in S$, there exists $y \in S$ such that $x < y$.

Therefore, S is a Dedekind cut.

Now, we consider the other possible signs of A and B. By 4.9, if C is a Dedekind cut, then $-C$ is also a Dedekind cut. If $A < \widehat{0}$ and $B \geq \widehat{0}$, then $-A \geq \widehat{0}$. By the nonnegative case, $(-A) \cdot B$ is a Dedekind cut. Therefore $A \cdot B = -((-A) \cdot B)$ is also a Dedekind cut. If $A \geq \widehat{0}$ and $B < \widehat{0}$, then $-B \geq \widehat{0}$. By the nonnegative case, $A \cdot (-B)$ is a Dedekind cut. Therefore $A \cdot B = -(A \cdot (-B))$ is also a Dedekind cut. Finally, if $A < \widehat{0}$ and $B < \widehat{0}$, then $-A \geq \widehat{0}$ and $-B \geq \widehat{0}$. By the nonnegative case, $(-A) \cdot (-B)$ is a Dedekind cut. Therefore, $A \cdot B = (-A) \cdot (-B)$ is a Dedekind cut. Hence, the product of any two Dedekind cuts is a Dedekind cut.

□

Definition 4.13. If $A > 0$, define

$$A^{-1} = \{r \in \mathbb{Q} : r < 0 \text{ or there exists } s \notin A \text{ with } s > 0 \text{ and } r < 1/s\}.$$

If $A < 0$, define

$$A^{-1} = -\big((-A)^{-1}\big).$$

The cut 0 has no reciprocal.

Proposition 4.14. *If $A \neq 0$, then A^{-1} is a cut and $AA^{-1} = \widehat{1}$.*

Proof. It is suffice to prove the result for $0 < A$, since the case $A < 0$ follows by applying positive case to $-A$ and then using the product sign rule.

First, we prove that A^{-1} is a Dedekind cut. We start with (C1). It is nonempty because it contains every negative rational. $A^{-1} \neq \mathbb{Q}$. Choose $0 < a \in A$. If $s \notin A$, then by 4.5, $a < s$. So, $\frac{1}{s} < \frac{1}{a}$. Therefore, every element in A^{-1} is less than $\frac{1}{a}$. So, $A^{-1} \neq \mathbb{Q}$. (C2) follows immediately. Finally, we consider (C3). If $r \in A^{-1}$ and r is negative, just take any positive element in A^{-1}. If $r \in A^{-1}$ and nonnegative, then $r < \frac{1}{s}$ for some $0 < s$ and $s \notin A$. Just consider $b = \frac{r+\frac{1}{s}}{2}$. $r < b < \frac{1}{s}$; thus, $b \in A^{-1}$ and (C3) is proven. Hence, A^{-1} is a Dedekind cut.

Now we prove $AA^{-1} = \widehat{1}$. We do this by proving $AA^{-1} \subseteq \widehat{1}$ and $\widehat{1} \subseteq AA^{-1}$.

$AA^{-1} = \left\{r \in \mathbb{Q} : r < 0 \text{ or there exist } a \in A,\ a' \in A^{-1} \text{ with } a > 0,\ a' > 0,\ r < aa'\right\}.$

First, we show $AA^{-1} \subseteq \widehat{1}$. If $x \in AA^{-1}$, and $x < 0$ then $x \in \widehat{1}$. Otherwise, $x < aa'$ for some $a \in A$ and $a' \in A^{-1}$. Also, since $a' \in A^{-1}$, there exists $0 < s \notin A$ and $a' < 1/s$ which means $s < \frac{1}{a'}$. For all $a \in A$, $a < s$ by 4.5. Thus, $\frac{1}{s} < \frac{1}{a}$. Multiplying the two inequalities, we have $1 < \frac{1}{aa'} \implies aa' < 1$ so we are done.

Now, we show that $\widehat{1} \subseteq AA^{-1}$. If $x \in \widehat{1}$ and $x < 0$ then $x \in AA^{-1}$ by definition. Otherwise, $0 \leq x < 1$. We need $a \in A$ and $a' \in A^{-1}$ such that $x < aa'$. Consider some number $c \notin A$ and then consider the set $\left\{cx^k : k \in \mathbb{Z}_{\geq 0}\right\}$.

Let $c' = x^k c \notin A$, where k is chosen so that $xc' = x^{k+1}c \in A$. Such a k exists because $x < 1$, as k increases in magnitude, cx^k decreases in magnitude. Indeed, write $x^{-1} = 1 + y$ with $y > 0$. By taylor series expansion, $x^{-n} = (1+y)^n \geq 1 + ny$, so $x^n \leq \frac{1}{1+ny}$. Choose some positive $a \in A$. For sufficiently large n, we have $\frac{c}{1+ny} < a$, and hence $cx^n < a$. Since A is downward closed, $cx^n \in A$. Choosing the smallest such n, we obtain $c' = x^{n-1}c \notin A$ while $xc' = x^n c \in A$.

Now we can choose $a > xc' \in A$ by (C3). So, $\frac{x}{a} < \frac{1}{c'}$. Next, we choose $a' = \frac{\frac{1}{c'}+\frac{x}{a}}{2}$ so that $\frac{x}{a} < a' < \frac{1}{c'}$. $a' \in A^{-1}$ since $c' \notin A$ and $a' < \frac{1}{c'}$. Thus, $x < aa'$; so, $x \in AA^{-1}$.

□

4.4. Ring Axioms

Proposition 4.15. *If A is a Dedekind cut, $A + (-A) = \widehat{0}$.*

Proof. $A + (-A) = \{a + a' : a \in A, a' \in -A\}$. Every $a' \in -A$ is less then some $-r$ where $r \notin A$.

We prove $A + (-A) \subseteq 0$. Suppose $x \in A + (-A)$. Then $x = a + a'$. Since $a' \in -A$, there exists some $r \notin A$ such that $a' < -r$. By 4.5, $a < r$. Therefore $a - r < 0$. So, $a + a' < a - r < 0$. Hence $x \in \widehat{0}$.

Now, we prove $\widehat{0} \subseteq A + (-A)$. Take $x \in \widehat{0}$. We want to choose an $a \in A$ so that $a' = x - a$ is in $-A$. For $x - a \in -A$, we need $r \notin A$ such that $x - a < -r$. So, $r - a < -x$. Since $x < 0$, we have $-x > 0$. So it is enough to find $a \in A$ and $r \notin A$ such that $r - a < -x$.

We claim that such a and r exist. Since $A \neq \mathbb{Q}$, choose some $r \notin A$. Since A is nonempty, choose some $a_0 \in A$. If $r - a_0 < -x$, then we are done. If not, then $r - a_0 \geq -x$. Since A has no greatest element, we can choose elements of A larger than a_0. Continue choosing larger elements until we get some $a \in A$ with $r - a < -x$. Then set $a' = x - a$.

Now $r - a < -x$, so $x < a - r$. This means $x - a < -r$. Since $r \notin A$, we have $a' = x - a \in -A$. Therefore $x = a + a'$, where $a \in A$ and $a' \in -A$. Hence $x \in A + (-A)$.

Thus $\widehat{0} \subseteq A + (-A)$. Therefore $A + (-A) = \widehat{0}$.

□

Proposition 4.16. *If A is a Dedekind cut, then $A + \widehat{0} = A$.*

Proof. We prove both containments. First, let $x \in A + \widehat{0}$. Then $x = a + b$ for some $a \in A$ and $b \in \widehat{0}$. Since $b \in \widehat{0}$, we have $b < 0$. Thus $x = a + b < a$. Since $a \in A$ and A is downward closed, $x \in A$. Therefore $A + \widehat{0} \subseteq A$.

Now let $x \in A$. Since A has no greatest element, there exists $y \in A$ such that $x < y$. Then $x - y < 0$, so $x - y \in \widehat{0}$. Since $x = y + (x - y)$, with $y \in A$ and $x - y \in \widehat{0}$, we have $x \in A + \widehat{0}$. Therefore $A \subseteq A + \widehat{0}$.

Hence $A + \widehat{0} = A$. □

Proposition 4.17. *If A is a Dedekind cut, then $A \cdot \widehat{1} = A$.*

Proof. First suppose $A \geq \widehat{0}$. We prove both containments. Let $x \in A \cdot \widehat{1}$. If $x < 0$, then $x \in \widehat{0} \subseteq A$. Otherwise, by the definition of multiplication, there exist $a \in A$ and $b \in \widehat{1}$ such that $a > 0$, $b > 0$, and $x < ab$. Since $b \in \widehat{1}$, we have $b < 1$. Thus $ab < a$. Since $a \in A$ and A is downward closed, $x \in A$. Therefore $A \cdot \widehat{1} \subseteq A$.

Now let $x \in A$. If $x < 0$, then $x \in A \cdot \widehat{1}$ because every negative rational is in the product of two nonnegative cuts. If $x \geq 0$, then by (C3), there exists $a \in A$ such that $x < a$. Choose a rational number b such that $0 < b < 1$ and $x < ab$. This is possible because $a > x \geq 0$, so we can choose $b < 1$ very close to 1. Since $b \in \widehat{1}$, we get $x \in A \cdot \widehat{1}$. Therefore $A \subseteq A \cdot \widehat{1}$.

So $A \cdot \widehat{1} = A$ when $A \geq \widehat{0}$. If $A < \widehat{0}$, then by the sign rule for multiplication, $A \cdot \widehat{1} = -((-A) \cdot \widehat{1})$. Since $-A > \widehat{0}$, the nonnegative case gives $(-A) \cdot \widehat{1} = -A$. Therefore $A \cdot \widehat{1} = -(-A) = A$.

Hence $A \cdot \widehat{1} = A$ for every Dedekind cut A. □

Proposition 4.18. *The other ring axioms.*

Proof. Ts trivial brochacho. Just think about it. □

4.5. Order Axioms

We show that the set of Dedekind cuts satisfies the following order axioms:

(D1) **Trichotomy:** For any cuts A and B, exactly one of the following holds: $A < B$, $A = B$, or $B < A$.

(D2) **Transitivity:** If $A < B$ and $B < C$, then $A < C$.

(D3) **Compatibility with Addition:** If $A < B$, then $A + C < B + C$.

(D4) **Compatibility with Multiplication:** If $A < B$ and $\widehat{0} < C$, then $AC < BC$.

Recall that the order is defined by inclusion: $A \leq B$ if and only if $A \subseteq B$, and $A < B$ if and only if $A \subset B$.

Proposition 4.19. *For any Dedekind cuts A and B, exactly one of the following holds: $A < B$, $A = B$, or $B < A$.*

Proof. If $A = B$, then we are done. Now suppose $A \neq B$. Then there is some rational number that belongs to one of the cuts but not the other.

First suppose there exists $a \in A$ such that $a \notin B$. We claim that $B \subseteq A$. Let $b \in B$. Since $a \notin B$, Lemma 4.5 applied to the cut B gives $b < a$. Since $a \in A$ and A is downward closed, we get $b \in A$. Therefore $B \subseteq A$. Also, since $a \in A$ but $a \notin B$, we know $A \neq B$. Hence $B \subset A$, so $B < A$.

Now suppose instead there exists $b \in B$ such that $b \notin A$. By the same argument, every element of A is less than b, so every element of A belongs to B. Therefore $A \subseteq B$. Since $b \in B$ but $b \notin A$, we have $A \neq B$. Hence $A \subset B$, so $A < B$.

Thus at least one of $A < B$, $A = B$, or $B < A$ holds. They cannot hold at the same time because equality and proper containment are mutually exclusive, and we cannot have both $A \subset B$ and $B \subset A$. Therefore exactly one holds. □

Proposition 4.20. *For any Dedekind cuts A, B, C, if $A < B$ and $B < C$, then $A < C$.*

Proof. Since $A < B$, we have $A \subset B$. Since $B < C$, we have $B \subset C$. Therefore, $A \subseteq C$. Thus, $A \subset C$, which means $A < C$. □

Proposition 4.21. *For any Dedekind cuts A, B, C, if $A < B$, then $A + C < B + C$.*

Proof. Since $A < B$, we have $A \subset B$. Let $x \in A + C$. Then $x = a + c$ for some $a \in A$ and $c \in C$. Since $A \subset B$, we have $a \in B$. Therefore $x = a + c \in B + C$. Hence $A + C \subseteq B + C$.

Now we show that the containment is proper. If $A+C = B+C$, then adding $-C$ to both sides gives $(A + C) + (-C) = (B + C) + (-C)$. By associativity and $C + (-C) = \widehat{0}$, this gives $A + \widehat{0} = B + \widehat{0}$, so $A = B$, contradicting $A < B$. Therefore $A + C \subset B + C$, so $A + C < B + C$. □

Lemma 4.22. *If $D > \widehat{0}$ and $E > \widehat{0}$, then $DE > \widehat{0}$.*

Proof. Since $D > \widehat{0}$, D contains a rational number d with $d \geq 0$. Since D has no greatest element, we may choose $d' \in D$ with $d < d'$. Then $d' > 0$. Similarly, since $E > \widehat{0}$, there exists $e' \in E$ with $e' > 0$. By the definition of multiplication, every negative rational is in DE, and since $0 < d'e'$, we also have $\frac{d'e'}{2} \in DE$. Therefore DE contains all negative rationals and at least one positive rational. Hence $\widehat{0} \subset DE$, so $DE > \widehat{0}$. □

Proposition 4.23. *For Dedekind cuts A, B, C, if $A < B$ and $\widehat{0} < C$, then $AC < BC$.*

Proof. Since $A < B$, by compatibility with addition we may add $-A$ to both sides to get $A+(-A) < B+(-A)$. Thus $\widehat{0} < B - A$. Since $\widehat{0} < C$, the previous lemma gives $(B - A)C > \widehat{0}$. Now add AC to both sides. This gives $AC < AC+(B-A)C$. By distributivity, $AC+(B-A)C = (A+B-A)C = BC$. Therefore $AC < BC$. □

4.6. Least Upper Bound

Proposition 4.24. *Let $\mathcal{A}$ be a nonempty set of Dedekind cuts. If $\mathcal{A}$ is bounded above, then $\mathcal{A}$ has a least upper bound.*

Proof. We claim that the union of the Dedekind cuts, $L = \bigcup_{A \in \mathcal{A}} A$, is the LUB of $\mathcal{A}$.

First, we prove that L is a Dedekind cut.

We look at (C1). L is nonempty since the Dedekind cuts in $\mathcal{A}$ are nonempty and we are taking the union of them. Now, we show that $L \neq \mathbb{Q}$. Since $\mathcal{A}$ is bounded above, $\exists U$, a Dedekind cut, such that $A \subseteq U$ for all $A \in \mathcal{A}$. Thus, we may choose $a \notin U$. Since $\mathcal{A} \subseteq U$ that means $L \subseteq U$; so, $a \notin L$. Thus, L does not contain every element in $\mathbb{Q}$; so, $L \neq \mathbb{Q}$.

Now, we look at (C2). Suppose we have $a \in L$ and $y < a$. We know that $a \in A_k$ for some $A_k \in \mathcal{A}$ by the definition of L. By (C2), this means that $y \in A_k$. Thus, $y \in L$.

Finally, we look at (C3). Suppose we have $a \in L$. This means $a \in A_k$ for some $A_k \in \mathcal{A}$. We want to show that there exists $b \in L$ such that $a < b$. By (C3), there exists $b' \in A_k$ such that $a < b'$. By the definition of L, $b' \in L$. Now, we just take $b = b'$.

Now, we prove that L is the LUB of $\mathcal{A}$.

First, L is an upper bound of $\mathcal{A}$ because every $A \in \mathcal{A}$ is contained in the union. Every element $a \in A$ is in L.

Now if M is any other upper bound, for $A \in \mathcal{A}$, $A \subseteq M$. So, their union $L \subseteq M$. Thus, L is the LUB.

□

5. Real Numbers

5.1. Construction

Definition 5.1. Define $\mathbb{R} = \{A \subseteq \mathbb{Q} : A \text{ is a Dedekind cut}\}$.

Definition 5.2. For $A, B \in \mathbb{R}$, define $A \leq B$ if $A \subseteq B$, and define $A < B$ if $A \subset B$.

Definition 5.3. We embed $\mathbb{Q}$ into $\mathbb{R}$ by sending each $q \in \mathbb{Q}$ to the rational cut $\widehat{q} = \{r \in \mathbb{Q} : r < q\}$.

Proposition 5.4. *Every nonempty subset of $\mathbb{R}$ with an upper bound has a least upper bound.*

Proof. Let $\mathcal{A}$ be a nonempty subset of $\mathbb{R}$ with an upper bound. Since $\mathbb{R}$ is the set of all Dedekind cuts, $\mathcal{A}$ is a nonempty set of Dedekind cuts with an upper bound. By the LUB proposition proved above for sets of Dedekind cuts, $\mathcal{A}$ has a least upper bound. Therefore every nonempty subset of $\mathbb{R}$ with an upper bound has a least upper bound. □

Proposition 5.5. $\mathbb{R}$ *is an ordered field.*

Proof. By definition, the elements of $\mathbb{R}$ are Dedekind cuts. Addition, additive inverses, multiplication, and multiplicative inverses were defined above for Dedekind cuts, and we proved that these

operations produce Dedekind cuts. Thus, these operations are well-defined on $\mathbb{R}$.

The additive identity is $\widehat{0}$, and the multiplicative identity is $\widehat{1}$. The propositions above show that $A + \widehat{0} = A$, $A + (-A) = \widehat{0}$, and $A \cdot \widehat{1} = A$. For $A \neq \widehat{0}$, the reciprocal A^{-1} was defined above and satisfies $AA^{-1} = \widehat{1}$. The remaining field axioms follow from the corresponding properties of rational addition and multiplication, since the operations on cuts were defined using rational sums and rational products. Therefore $\mathbb{R}$ is a field.

The order on $\mathbb{R}$ is defined by inclusion. In the order section above, we proved trichotomy, transitivity, compatibility with addition, and compatibility with multiplication by positive elements. Therefore $\mathbb{R}$ is an ordered field. □

5.2. Uniqueness

We now prove the uniqueness of $\mathbb{R}$. More precisely, suppose F is an ordered field with the least upper bound property, and suppose $\mathbb{Q}$ embeds into F. We show that F is isomorphic to $\mathbb{R}$.

We define a map $\phi : F \to \mathbb{R}$ by $\phi(x) = \{q \in \mathbb{Q} : q < x\}$. Here each rational number q is viewed as its copy inside F. In words, $\phi(x)$ is the set of all rational numbers lying to the left of x.

First, we show that $\phi(x)$ is a Dedekind cut. It is nonempty and not all of $\mathbb{Q}$ because there are rational numbers less than x and rational numbers greater than x. It is downward closed because if $q \in \phi(x)$ and $r < q$, then $r < x$, so $r \in \phi(x)$. It has no greatest element because if $q < x$, then there is a rational number s such that $q < s < x$, so $s \in \phi(x)$. Therefore $\phi(x) \in \mathbb{R}$.

Next, ϕ preserves order. If $x < y$, then every rational number less than x is also less than y, so $\phi(x) \subseteq \phi(y)$. Also, there is a rational number q such that $x < q < y$. Then $q \in \phi(y)$ but $q \notin \phi(x)$, so $\phi(x) \subset \phi(y)$. Hence $x < y$ implies $\phi(x) < \phi(y)$.

This also shows that ϕ is injective. Indeed, if $x \neq y$, then either $x < y$ or $y < x$. In either case, the previous paragraph shows that $\phi(x) \neq \phi(y)$.

Now we show that ϕ is surjective. Let $A \in \mathbb{R}$. Since A is a Dedekind cut, it is nonempty and bounded above in $\mathbb{Q}$, and hence also bounded above in F. Since F has the least upper bound property, A has a least upper bound in F. Let $x = \sup A$. We claim that $\phi(x) = A$.

First, suppose $q \in A$. Since A has no greatest element, there exists $a \in A$ such that $q < a$. Since x is an upper bound for A, we have $a \leq x$. Therefore $q < x$, so $q \in \phi(x)$. Hence $A \subseteq \phi(x)$.

Conversely, suppose $q \in \phi(x)$. Then $q < x$. Since $x = \sup A$, the rational number q cannot be an upper bound for A. Therefore there exists some $a \in A$ such that $q < a$. Since A is downward closed, $q \in A$. Hence $\phi(x) \subseteq A$. Therefore $\phi(x) = A$. So ϕ is surjective.

Finally, ϕ preserves addition and multiplication because both operations on cuts were defined using rational sums and rational products. More specifically, $\phi(x+y) = \phi(x)+\phi(y)$ and $\phi(xy) = \phi(x)\phi(y)$, with the multiplication statement following from the same sign cases used in the definition of multiplication of cuts. Also, $\phi(0_F) = \widehat{0}$ and $\phi(1_F) = \widehat{1}$.

Thus ϕ is a bijection that preserves addition, multiplication, and order. Therefore F is isomorphic to $\mathbb{R}$ as an ordered field. This proves that the real numbers are unique up to ordered-field isomorphism.

6. Cauchy Sequences

6.1. Completeness

Definition 6.1. A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is called *Cauchy* if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that whenever $n, m > N$, we have $|a_n - a_m| < \epsilon$.

Definition 6.2. A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ *converges* to $a \in \mathbb{R}$ if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that whenever $n > N$, we have $|a_n - a| < \epsilon$.

Proposition 6.3. *Every Cauchy sequence in* $\mathbb{R}$ *converges.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{R}$. We want to show that the sequence converges to some $a \in \mathbb{R}$.

Let $S = \{x \in \mathbb{R} : \text{ there exists } N \in \mathbb{N} \text{ such that for all } n > N,\ x < a_n\}$. In words, S is the set of real numbers that are eventually less than every term of the sequence.

First, we show that S is nonempty. Since (a_n) is Cauchy, take $\epsilon = 1$. Then there exists $N_0 \in \mathbb{N}$ such that whenever $n, m > N_0$, we have $|a_n - a_m| < 1$. Choose some fixed $k > N_0$. Then for every $n > N_0$, we have $|a_n - a_k| < 1$, so $a_k - 1 < a_n < a_k + 1$. Therefore $a_k - 1$ is eventually less than every a_n, so $a_k - 1 \in S$. Hence $S \neq \varnothing$.

Now we show that S is bounded above. We claim that $a_k + 1$ is an upper bound for S. Let $x \in S$. By definition of S, there exists $N_1 \in \mathbb{N}$ such that for all $n > N_1$, we have $x < a_n$. Choose $n > \max(N_0, N_1)$. Then $x < a_n$ and $a_n < a_k + 1$. Therefore $x < a_k + 1$. Hence every $x \in S$ is less than $a_k + 1$, so S is bounded above.

Since S is nonempty and bounded above, the least upper bound property gives us a number $a = \sup S$. We claim that $a_n \to a$.

Let $\epsilon > 0$. Since (a_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that whenever $n, m > N_0$, we have $|a_n - a_m| < \epsilon/2$. Choose some fixed $k > N_0$. Then for every $n > N_0$, we have $a_k - \epsilon/2 < a_n < a_k + \epsilon/2$. Therefore $a_k - \epsilon/2 \in S$. Since $a = \sup S$, this gives $a_k - \epsilon/2 \leq a$.

We also claim that $a_k + \epsilon/2$ is an upper bound for S. Let $x \in S$. Then there exists $N_1 \in \mathbb{N}$ such that for all $n > N_1$, we have $x < a_n$. Choose $n > \max(N_0, N_1)$. Then $x < a_n$ and $a_n < a_k + \epsilon/2$. Therefore $x < a_k + \epsilon/2$. So $a_k + \epsilon/2$ is an upper bound for S. Since a is the least upper bound of S, we get $a \leq a_k + \epsilon/2$.

Thus $a_k - \epsilon/2 \leq a \leq a_k + \epsilon/2$. For every $n > N_0$, we also have $a_k - \epsilon/2 < a_n < a_k + \epsilon/2$. So both a and a_n lie within the interval from $a_k - \epsilon/2$ to $a_k + \epsilon/2$, whose length is ϵ. Therefore $|a_n - a| < \epsilon$ for all $n > N_0$. Hence $a_n \to a$.

Thus every Cauchy sequence in $\mathbb{R}$ converges. □

7. Conclusion

We constructed the real numbers from the rational numbers by defining $\mathbb{R}$ to be the set of all Dedekind cuts. Each rational number q is represented by the rational cut $\widehat{q} = \{r \in \mathbb{Q} : r < q\}$, while irrational numbers correspond to cuts that do not arise from any rational boundary. In this way, Dedekind cuts fill the gaps in $\mathbb{Q}$ using only subsets of rational numbers.

We defined addition, additive inverses, multiplication, and multiplicative inverses for Dedekind cuts and showed that these operations make $\mathbb{R}$ into a field. We also defined the order on $\mathbb{R}$ by inclusion and verified that it is compatible with the field operations. Therefore, $\mathbb{R}$ is an ordered field containing an isomorphic copy of $\mathbb{Q}$.

Most importantly, we proved that every nonempty collection of Dedekind cuts that is bounded above has a least upper bound, given by the union of the cuts in the collection. Thus, $\mathbb{R}$ satisfies the least upper bound property and is therefore a complete ordered field. This completeness fills the gaps present in $\mathbb{Q}$ and guarantees the existence of numbers such as $\sqrt{2}$.

Finally, the uniqueness theorem shows that any complete ordered field containing $\mathbb{Q}$ is isomorphic to the real numbers constructed here. The least upper bound property also implies that every Cauchy sequence in $\mathbb{R}$ converges. Hence, the Dedekind cut construction produces, up to isomorphism, the unique complete ordered field that we call the real numbers.